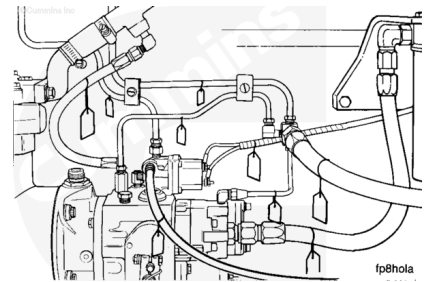


Trainings ▾

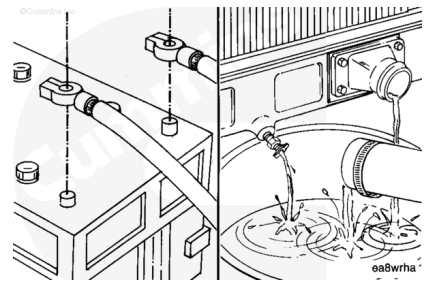
Remove

Put a tag on all hoses, lines, linkage, and electrical connections as they are removed to identify location and aid during the installation process.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

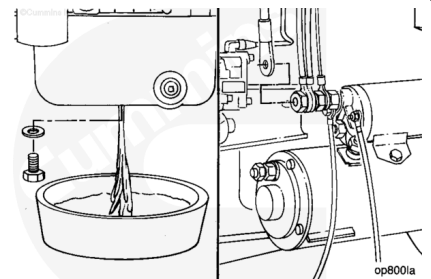


Disconnect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).

Drain the engine coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



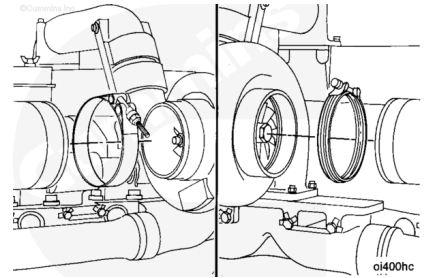
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

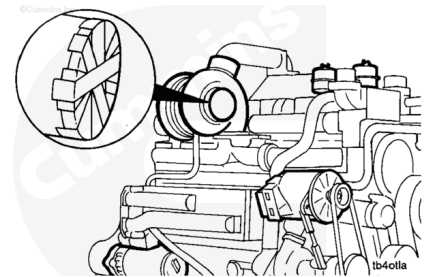
Drain the lubricating oil.

Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wires and hydraulic lines.

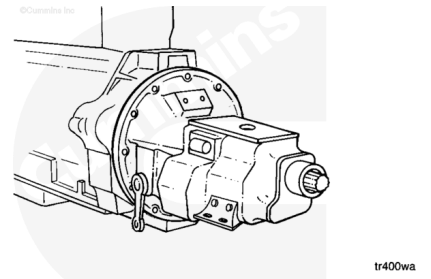
Disconnect the intake and exhaust system pipes.



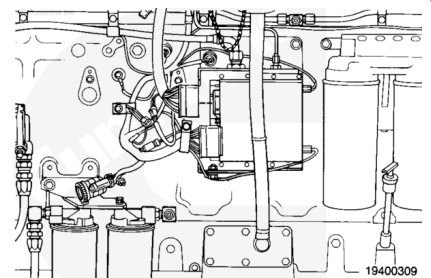
Cover all engine openings to prevent dirt and debris from entering the engine.



Disconnect the driven units from the flywheel housing and flywheel.

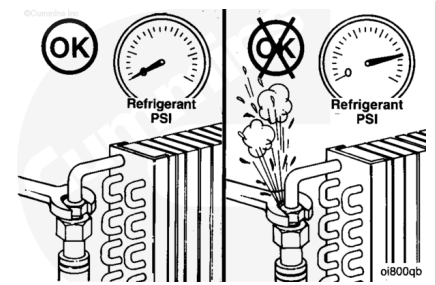


Disconnect the OEM wiring harness. It is located at a junction connection at the rear of the fuel pump side of the engine.
Refer to Procedure 019-031
(</qs3/pubsys2/xml/en/procedures/13/13-019-031.html>) in the Electronic Control System QSK19 CM850, Modular Common Rail System, Bulletin 4021493.



⚠ WARNING ⚠

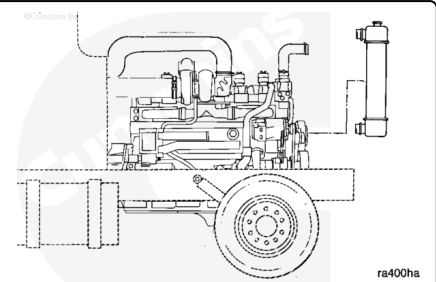
If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fitting before removing. Liquid refrigerant can cause serious eye and skin injuries.



⚠ WARNING ⚠

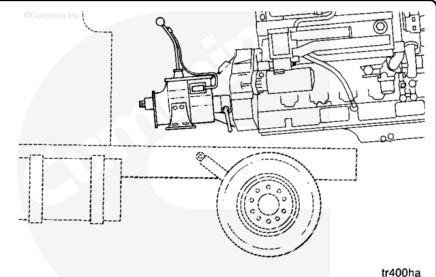
To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

Remove all chassis components necessary to remove the engine from the equipment.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

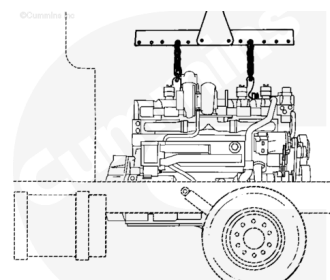
Note : On applications when the rear engine mounts are attached to the driven unit, it is sometimes necessary to remove the engine and driven unit as an assembly.

Refer to the OEM troubleshooting and repair manual for the weight of the drive unit. The engine lifting fixture, Part Number 3162871, is designed to lift a maximum of 2722 kg [6000 lb].

Use a properly rated hoist and engine lifting fixture, Part Number 3162871, to remove the engine and to support the weight of the engine. Remove the front and rear engine support capscrews.

Note : If the driven unit is not removed, put a support under the driven unit to prevent it from falling.

Use 2 Lifting Hooks, Part Number 3163091, to lift the engine. Place the hooks between the cylinder heads, under the lifting lugs.

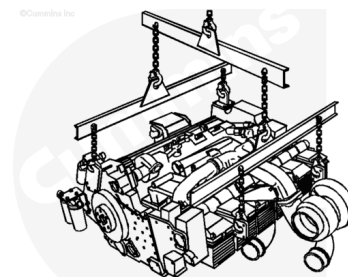


ew400ha

Rail Applications

⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



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⚠ WARNING ⚠

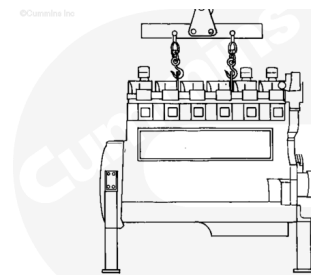
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Install engine lifting fixture, Part Number 3162871, to engine lifting brackets as shown. (3 lifting fixtures required)

Remove the engine.

⚠ WARNING ⚠

Use a stand or skid that contacts the engine mounts. The stand must hold the weight of the engine and provide permanent support to prevent the engine from falling.



ew4suha

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

The oil sump or cover plate will not support the weight of the engine. Damage to the engine can occur.

Put the engine on the stand.

Remove all remaining accessories and brackets that will be used on the replacement engine.

